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Chinook and Summer Chum Salmon Abundance, Run Timing, and Age, Sex, and Length Composition in the East Fork of the Andreafsky River, Yukon Delta National Wildlife Refuge, Alaska, 2021

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Cover Photo: Aerial view of the East Fork Andreafsky River weir, 2021. Photo courtesy of S. Ransbury, USFWS.

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Abstract

A resistance board weir was operated on the East Fork Andreafsky River from June 18 through July 28, 2021, to estimate escapement and run timing of Chinook *Oncorhynchus tshawytscha* and summer Chum *O. keta* salmon. Other fish species passing the weir were counted incidentally. Fish were identified to species and counted from video recordings generated by an underwater video camera and recorder. A highwater event necessitated that part of the weir be removed from 2000 hours on July 9 to 1130 hours on July 12, leading to reduced counts during that period. Fish passage at the tails of the run were within the historically known migration period, but before and after weir operation, were estimated using a Bayesian statistical procedure. Chinook and summer Chum salmon were sampled to estimate age, sex, and length composition of the escapement. The Chinook Salmon count was 1,425 fish, and total estimated escapement was 1,454 fish (95% Bayesian highest posterior density interval, HPDI, 1,435–1,478 fish). The summer Chum Salmon count was 2,534 fish, and total estimated escapement was 2,684 fish (95% HPDI 2,596–2,772 fish). Water temperatures were low enough to allow for sampling during the entire season, but low escapement numbers of target species compared to other species made it difficult to collect sample sizes for Chinook Salmon. Only 89 useable Chinook Salmon scale samples were collected, primarily from the latter half of the run. Therefore, the samples were insufficient to characterize the age and sex composition of the overall escapement. However, the Chinook Salmon scales that were collected were 42% age-4, 38% age-5, and 19% age-6, plus 1% age-2.1 jacks. The Chinook Salmon samples were 31.5% female and 68.5% male. The 224 valid summer Chum Salmon age samples were sufficient to characterize escapement age and sex composition as 86% age-0.3, 10% age-0.4, 3% age-0.5, and 1% age-0.2. Females were more numerous, at 56% of escapement, than males (44%). Incidental counts of other species included 7,030 Pink Salmon *O. gorbuscha*, 241 Sockeye Salmon *O. nerka*, 115 Northern Pike *Esox lucius*, and 3,544 whitefish (Coregoninae).

Introduction

The Andreafsky River is a major salmon producing tributary of the lower Yukon River, and supports some of the largest runs of Chinook *Oncorhynchus tshawytscha* and summer Chum *O. keta* salmon in the Yukon River drainage. Located within the Yukon Delta National Wildlife Refuge, the Andreafsky River is designated a wild and scenic river. The Andreafsky River is also the lowest major tributary in the Yukon River drainage, entering the Yukon River main stem about 160 river kilometers (rkm) from the Bering Sea. The Yukon River extends over 3,190 rkm from ice fields in British Columbia, Canada near the Gulf of Alaska, through the Yukon Territory and across interior Alaska, to its delta on the Bering Sea coast. The escapement

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monitoring project on the East Fork of the Andreafsky River is a key location in a drainage-wide network of salmon monitoring and assessment projects, representing lower river stocks and providing information on salmon abundance, run timing, and stock composition. Escapement estimates from the East Fork Andreafsky River weir are combined with aerial survey estimates from both the East and West forks of the Andreafsky River and are used by fisheries managers as in-season and post-season indices of lower river escapement.

Subsistence fisheries in the U.S. portion of the Yukon River drainage are jointly managed by the State of Alaska, through the Department of Fish and Game (ADF&G), and the Federal Subsistence Management program, through the U.S. Fish and Wildlife Service (USFWS). Commercial, sport, and personal use fisheries are managed by ADF&G. Within the National Wildlife Refuge, federal managers are responsible for conserving fish and habitats in their natural diversity, supporting subsistence resource use by local residents, and protecting water quality (Yukon Delta National Wildlife Refuge Resource Management). Both state and federal managers are responsible for ensuring priority of subsistence fishing use, protecting stock diversity, ensuring equitable distribution of harvest opportunity across the drainage, and meeting Canadian border passage objectives as specified by the Yukon River Salmon Agreement[a]. In addition, ADF&G establishes escapement goals for various salmon populations in Yukon River tributaries, including East Fork Andreafsky River Chinook and summer Chum salmon (Volk et al. 2009; Fleischman and Evenson 2010). At the East Fork Andreafsky weir, Chinook Salmon have a sustainable escapement goal (SEG) of 2,100-4,900 fish and summer Chum Salmon have an SEG of >40,000 fish (JTC 2021).

Salmon returning to the Andreafsky River contribute substantially to subsistence and commercial harvests in the lower Yukon River. Residents of 20 communities located on the Yukon Delta and Yukon River below the Andreafsky River confluence harvest salmon from mixed stocks entering the Yukon River. Most residents harvest fish for subsistence use, but many also participate in small-scale commercial fisheries. In the mixed cash-subsistence economy typical of the region, income from commercial fishing enables families to purchase the necessary gear and supplies to sustain their subsistence fishing activities. Chinook Salmon are the primary target for subsistence fisheries in the lower river, but summer and fall Chum Salmon and Coho Salmon (*O. kisutch*) are also harvested. Commercial fisheries in the lower river typically harvest summer Chum Salmon during the early season, and shift to fall Chum and Coho salmon after mid-July.

Most subsistence and commercial fishing in the lower Yukon River occur in the main stem, harvesting from mixed stocks of salmon returning to spawning areas throughout the drainage. Although individual stocks cannot be specifically targeted or avoided, in-season genetic information and daily updates from escapement counting projects are monitored closely to ensure that certain stocks are not disproportionately harvested. Genetic mixed-stock analysis (MSA) compares genotypes of harvested fish with a baseline of genotypes from known spawning populations. Genetic baselines have been developed for both Chinook and Chum salmon[b] (Flannery et al. 2007) in the Yukon River. For each species, lower river spawning populations, occurring downstream from the main stem just above the delta to the Gisasa River in the lower Koyukuk River drainage, are genetically similar and grouped together as the lower Yukon River stock. All populations within the lower Yukon River Chum Salmon stock are genetically identifiable as summer Chum Salmon. MSA sampling, however, occurs at the Pilot Station sonar project and does not include lower river populations such as those in the

[a] https://www.yukonriverpanel.com/publications/yukon-river-salmon-agreement
[b] http://www.adfg.alaska.gov/index.cfm?adfg=fishinggeneconservationlab.yukonchinook_baseline

Andreafsky River, which spawn below the sonar site. Therefore, direct monitoring projects are needed to estimate the complete contribution of lower river populations to the mixed-stock runs. Furthermore, because MSA does not capture fine-scale distinctions among populations within the lower Yukon River Chinook and summer Chum salmon stocks, escapement monitoring projects such as the East Fork Andreafsky River weir help protect geographically distinct populations.

The Andreafsky River has been an important indicator for lower Yukon salmon runs for many years. Aerial surveys were conducted on both the East and West forks in most years since 1954[c]. ADF&G experimented with sonar on the East Fork in the early 1980s (Buklis 1982) and operated a fish counting tower there from 1986 to 1988. The East Fork Andreafsky River weir in its current configuration has been operated annually by USFWS since 1994; except in 2020 when operation was suspended due to the Covid-19 pandemic. Escapement information from East Fork Andreafsky River is useful for in-season management, lagging run timing through the lower river fisheries by only a few days.

Similar to previous years, daily migration counts of all fish species, and sex, length, and scale samples from Chinook and summer Chum salmon were collected at the East Fork Andreafsky River weir in 2021. Water depth, temperature, and weather conditions were also monitored. The underwater video system incorporated into the primary fish passage chute in 2014 continues to be used for counting fish passage through the weir. The 2021 season continued to use the newer style of weir panels manufactured before the 2019 season.

The following objectives were established for the 2021 season:

1. Estimate daily and seasonal escapement and run timing of adult Chinook and summer Chum salmon.
2. Estimate the age, sex, and length composition of the adult Chinook and summer Chum salmon escapements, with 95% confidence intervals for age-sex proportions no larger than ± 0.1.
3. Identify and count other fish species passing through the weir daily and collect water level (twice daily) and temperature (every 15 minutes) data.

Study Area

The project site is on the East Fork of the Andreafsky River, about 40 rkm upriver from the village of St. Mary's (Figure 1). St. Mary's is on the main-stem Andreafsky River, near its confluence with the Yukon River. The upland area surrounding the weir site, including the weir materials storage area and campsite, are on Nerklikmute Native Corporation land which is leased to the U.S. Fish and Wildlife Service for the project. The weir site within the channel of the East Fork Andreafsky River is managed by the State of Alaska Department of Natural Resources, which issues a Land Use Permit[d] to USFWS for the weir.

The Andreafsky River is in the lower reaches of the Yukon River drainage, flowing into the main-stem Yukon River about 160 rkm upriver from the Bering Sea. Its headwaters in the Nulato Hills give rise to two roughly parallel tributaries (East and West forks) which flow in a

[c] http://www.adfg.alaska.gov/CommFishR3/Website/AYKDBMSWebsite/Default.aspx
[d] http://dnr.alaska.gov/mlw/forms/#land+land-use-authorization

southwesterly direction for over 200 rkm before converging about 7 rkm upstream of the confluence with the Yukon River. The total watershed area is approximately 5,450 km^2. The Andreafsky River supports runs of Chinook, summer Chum, Pink *O. gorbuscha*, Sockeye *O. nerka*, and Coho salmon, and is one of the largest salmon-producing tributaries in the lower Yukon River. Almost all salmon fishing on these stocks occurs below the confluence of the East and West forks and in the Yukon River. Other anadromous fish present in the Andreafsky River include Humpback *Coregonus clupeaformis*, Broad *C. nasus*, and Round *Prosopium cylindraceum* whitefish; Sheefish *Stenodus leucichthys*; Least *C. Sardinella* and Bering *C. Laurettae* cisco; and Dolly Varden *Salvelinus malma*. Resident freshwater species include Arctic Grayling *Thymallus arcticus*, Northern Pike *Esox lucius*, Burbot *Lota lota*, and Longnose Sucker *Catostomus catostomus*.

The East Fork Andreafsky River flows from its source for approximately 50 rkm through alpine tundra, and then for approximately 130 rkm through a forested river valley bordered by hills mostly less than 400 m in elevation, with willow, spruce, alder, and birch dominating the riparian zone and much of the hillsides. The streambed in this section is characterized by glides and riffles with gravel and rubble substrate. The river widens in its lowermost portion, flowing through wetlands, interspersed with forest and tundra, and bordered by hills that are typically less than 230 m in elevation. In this low gradient section of the river, water levels can be affected by fluctuations on the Yukon River. The regional climate is subarctic. Temperature records from St. Mary's for the most recent 10-year period (2011–2020) show an average summer temperature (Jun-Aug) of 11.3°C and an average winter temperature (Dec-Feb) of -10.9°C[e].

Methods

Weir Design and Operation

The weir structure is a modified resistance board design (Tobin 1994; Tobin and Harper 1995) with 3.5 cm pickets with a 4.8 cm gap between each picket. This spacing allows smaller Pink Salmon and resident fish to pass through the weir undetected, reducing counts of those non-target species, and allows for better water flow through the weir. The weir site coordinates were N 62° 07', W 162° 48.4', and the channel width is 105 m at that point. The main fish passage chute was located at the deepest part of the channel, approximately 20 m from the left bank, and directed fish into the sampling trap and video chute. The video chute contained a camera box with a glass view window and a CAM-AM070 Color Analog Video Camera (Applied MicroVideo). Video recording of all fish within the video chute was activated by motion capture software (Security Spy) on a computer system housed in the main cabin and connected by a cable to the camera at the weir. Recorded video files were replayed later for species identification and enumeration.

The weir and video system are normally operated 24 hours a day, 7 days a week, except for possible periods of high water or other problems. The weir was inspected each day and was cleared of debris at least once every 8 hours. Minor repairs were made to the weir as needed to ensure that it remained structurally sound and provided a fish-tight barrier. The video system was set to record a motion capture video file every hour. Crew members replayed files recorded during their respective shifts to identify species and count all fish that passed through the video chute. Counts were tallied with a computer keypad directly into a Visual Basic for Applications

[e] Climate Data Online, National Oceanic and Atmospheric Administration, National Centers for Environmental Information, https://www.ncdc.noaa.gov/cdo-web/datasets#LCD, accessed 9/8/2021.

program in Excel, which automatically compiled a text file of fish counts for each hour and generated a 24-hour daily summary in Excel. Each morning, the previous day's data were reported by satellite phone to the Northern Alaska Fish and Wildlife Conservation Office and included in a daily escapement monitoring summary report distributed to managers, biologists, and stakeholders.

Quality Control Review of Species Identification and Passage Counts

Post-season, all hourly video files from selected days were reviewed as a quality control measure. The four days of highest fish passage were selected for review because counting and species identification errors would be more likely when fish were crowded in the passage chute and view window. Additional video files were reviewed as necessary to follow up on significant discrepancies between field and post-season counts. Fish identification discrepancies were assessed qualitatively to determine the nature of the errors, look for differences between individual observers, and track whether accuracy improved over time. Discrepancies in counts were also assessed quantitatively for presence of systematic bias (i.e., consistent under- or over-counting) and magnitude of error compared with total species counts.

Visual fish counts at weirs, whether by direct observation or through video recordings, are generally assumed to be a complete census of the species passage during the operating period, and error is assumed to be negligible. Therefore, formal variance statistics are not generally reported, and we do not report them in this project.

Computation of Passage for Non-Operational Days and Tails of the Run

The weir is seasonally installed within dates that are expected to observe greater than 95% of fish passage for the two target species, Chinook and summer Chum salmon. However, fish are known to pass in small numbers before and after the operational period of the weir (Brown et al. 2020). To estimate fish passage during the tails of the run, i.e., days within the typical migration period for each target species but before and after the weir operational period, a statistical arrival model was used (Sethi and Bradley 2016). This procedure was also used to estimate fish passage on days when fish were able to escape through the weir uncounted due to high water or other problems. However, if the weir remained functional and fish were counted through the passage chute for at least 6 hours in a given day, the actual count expanded for a 24 hour period was reported, and the estimate was not used. The model could potentially include low probability events of fish passage on days far outside the typical migration period, so it was constrained to a realistic range of dates. The range used was June 15 through August 10, based on historical Chinook and summer Chum salmon passage data from weir operations during 1994 through 2017.

When modeling arrival of fish at the weir, three possible distributions were considered: Normal, skew-Normal, and Student-*t*, with two degrees of freedom. Variability around this smooth arrival curve was modeled by Negative Binomial random variables, such that the magnitude of variability scaled positively with estimated daily passage counts. Three correlation structures were included for process variation in the Negative Binomial probability model: none (white noise), lag-1 autoregressive, and lag-1 moving average. All combinations of run shape and process variation structure were fit to the data, resulting in a candidate set of nine potential models. Model fits were made in a Bayesian framework, allowing predicted estimates for each missing passage date and the associated variability to be easily obtained. Prior distributions were specified (Table 1 in Sethi and Bradley 2016) and models were run in OpenBUGS (Spiegelhalter et al. 2007) using the R package R2OpenBUGS (Sturtz et al. 2019). Two chains were run with a burn-in period of 1,000 iterations and a thin rate of 10, resulting in 2,000 posterior draws stored. For each model, convergence was ensured by confirming that the Gelman-Brooks-Rubin statistic for the estimated total population abundance was less than 1.2.

Deviance information criteria (DIC) were used to select the best model from the candidate set of nine models. Predicted passage from the initial and concluding tails of the run was summed separately and reported as derived parameters. For days within the weir operation period, only the observed counts expanded for 24 hours were reported for days during which the weir was fully operational for at least 6 hours. If counting was conducted for less than 6 hours, model estimated passage was reported. The total escapement estimate was the sum of fish counted during the weir operation period, arrival model estimates for the tails of the run, and missed passage estimates on any days with less than 6 hours of counting. A 95% highest posterior density interval (HPDI), which is roughly a Bayesian equivalent of a 95% confidence interval, was formed by adding 2.5 and 97.5 percentiles of the posterior distribution to the total counted fish.

Biological Sampling and Data Analysis

Sex and length data and scale samples were collected from Chinook and summer Chum salmon in a trap connected to the main passage chute. Both ends of the trap were closed to retain fish during sampling but were open at all other times. Water temperature was monitored to ensure that trapped fish were not in danger of heat stress, with a goal to avoid trapping if the daily average water temperatures were above 17°C for three consecutive days or 20°C at any given moment (Shink 2020). The sample size goal for each species was 220–240 fish for the season. This goal was based on a statistical calculation indicating that a minimum of 180 samples are required to obtain eight simultaneous 95% confidence intervals for age-sex classes such that each interval is no more that ± 0.1 unit wide (Bromaghin 1993). The sample size was increased by 25% above this minimum as an allowance for unreadable scales and other sampling errors. To accurately characterize the composition of the entire run, the total number of samples must be distributed throughout the run roughly in proportion to escapement, while not exceeding the target number of samples. As a pre-season guideline for sampling, the season was divided into 3 periods or tertiles based on historical fish passage counts, and the target sample size within each tertile was 73–80 fish. The tertile dates designated for Chinook Salmon were June 16–July 8, July 9–14, and July 15–31. The tertile dates designated for summer Chum Salmon were June 16–July 3, July 4–10, and July 11–31. To ensure consistent sampling effort towards meeting the target sample size for each tertile, weekly and daily targets were set. Salmon were sampled opportunistically during a day or week until the daily target was met. For age determination, three scales from each Chinook Salmon and one scale from each summer Chum Salmon were collected from the preferred sampling area (INPFC 1963). Scales were mounted on gum cards matched with length and sex data from each fish and sent to the Stock Biology group at ADF&G Division of Commercial Fisheries in Anchorage for age analysis (Eaton 2015). Fish length was measured from mid-eye to fork of tail, and sex was visually determined or, in the case of Chinook Salmon, based on length. Based on historical sampling in the East Fork Andreafsky River, mid-eye to fork lengths (MEFL) of 650 mm form a reliable barrier between the smaller 2-ocean fish and the larger 3-ocean fish. Returning female Chinook Salmon in the 2-ocean age class, known as jills, are extremely rare, so all Chinook Salmon <650 mm were considered males (Brady 1983; Karpovich and DuBois 2007; Larson 2020). For Chinook Salmon ≥650 mm MEFL and all Chum Salmon, external characteristics such as kype development or the presence of an ovipositor were used for determining sex.

Because the designated tertiles based on previous years' run timing did not necessarily correspond to actual run timing, the samples were stratified post-season based on 2021 run timing. Weighted estimates from each tertile were then combined to form seasonal estimates of age and sex composition.

Means and standard errors were calculated and reported for fish lengths within each age and sex class. Lengths within each age-sex class were assumed not to vary substantially over the seasonal progression of the run, so length calculations were made on pooled, unweighted samples within each age-sex class.

Age and sex composition were calculated as population proportions based on the number of sampled fish in each age and sex class. In each sampling period i, a number of fish n_i were sampled, and of these a number n_{ij} were determined to be of class j (by age, sex, or age and sex combined). The proportion p_{ij} of fish of class j in sampling period i was estimated as:

$$\widehat{p_{ij}} = \frac{n_{ij}}{n_i}$$

The variance of the proportion of class j in period i was:

$$\widehat{V_{p_{ij}}} = \frac{n_i}{n_i - 1} \cdot \widehat{p_{ij}} \cdot \left(1 - \widehat{p_{ij}}\right)$$

Sample proportions were weighted by the proportion N_i/N of the total population N available for sampling within period i (i.e., proportion of the total seasonal escapement counted during period i). Weighted sample proportions were summed across K periods to provide a pooled seasonal estimate of the proportion of class j, as:

$$\widehat{p_j} = \sum_{i=1}^{K} \frac{N_i}{N} \cdot \widehat{p_{ij}}$$

The variance of each seasonal age, sex, or age-sex class proportion was estimated as:

$$\widehat{V_{p_j}} = \sum_{i=1}^{K} \left(1 - \frac{n_i}{N_i}\right)\left(\frac{N_i}{N}\right)^2 \cdot \frac{\widehat{p_{ij}}\left(1 - \widehat{p_{ij}}\right)}{n_i - 1}$$

Using the above equation for variance and assuming asymptotic normality, 95% confidence interval (CI) was calculated as:

$$\widehat{p_j} \pm 1.96 \cdot \sqrt{V_{p_j}\left(\widehat{p_j}\right)}$$

Other Species Counts and Environmental Data

Other salmon and non-salmon species were also recorded and counted using the video system. Daily passage counts of these other fish species were documented and reported in-season, but no corrections were made for missed passage before, after, or during the weir operation period.

Because counts for non-target species may not have encompassed the entire migration period of each species, they would not be considered reliable indicators of the species' run timing, migration behavior, escapement, or stock status. However, they do provide information about presence, absence, and co-migration with the target species.

Water depth was manually measured twice daily at approximately 0700 hours and 1900 hours near the fish passage chute, just upstream of the weir. A staff gauge was installed next to the sampling trap and used to measure water levels. The staff gauge was not calibrated to a fixed benchmark and thus, it only provided a relative measure of water depth during the season. However, in 2021 a benchmark was installed on the shore to standardize future staff gauge readings over the next few seasons. Water temperature was measured in continuous 15-minute increments with a Heron Dipper Log vented probe (Heron Instruments, Inc.) fixed next to the trap via buoy and anchor to maintain a constant water depth (0.3 m) and location. In addition to water depth and temperature, air temperature and other weather information were monitored and reported in the daily in-season summary.

Results

Weir Operation

Placement of the weir in 2021 was completed in the afternoon of June 17. Normal weir operations began at 0000 hours on June 18 and continued through 2359 hours on July 28 (Table 1). Flooding resulted in artificially low counts from 2000 hours on July 9 to 1130 hours on July 12, as the weir was not fish tight during that period.

Quality Control Review of Species Identification and Passage Counts

An analysis to determine the discrepancy of daily fish counts for Chinook and summer Chum salmon was conducted on the four highest days (July 17–July 20) of fish passage and compared with post-season counts. The discrepancy between the inseason daily count of Chinook Salmon and the post-season counts ranged from 0.3% (July 19) to 5.1% (July 17). Discrepancies in summer Chum Salmon between daily inseason counts compared with post-season counts ranged from 0.3% (July 19) to 2.4% (July 17). Observed differences were both positive and negative, indicating a lack of systematic or directional counting bias.

Arrival model estimates for escapement before the beginning (June 15–18) and after the end of weir operation (July 29–August 10) added approximately 29 Chinook Salmon and 150 summer Chum Salmon to the total passage counts from the weir. The best-fit model for both species was a Student-t, but Chinook Salmon data fit a lag-1 moving average model instead of a model with no correlation structure such as the one that summer Chum Salmon data fit best with. These model estimates suggest that few or no fish passed before weir operation began. Combining counts and estimates, the final escapement estimates were 1,454 Chinook Salmon (95% HPDI 1,435–1,478 fish) and 2,684 summer Chum Salmon (95% HPDI 2,596–2,772 fish; Table 1). Due to flooding during July 10–11, there were major holes in the weir where fish could pass uncounted for greater than 18 hours each day, so missed passage estimates were calculated from the best fit models. However, the estimates for those two days were ultimately not used because the fish that did continue to go through the counting shoot during that period numbered more than the estimates.

Chinook and Summer Chum Salmon Escapement and Age-Sex-Length Composition

During 2021 weir operations 1,425 Chinook Salmon and 2,534 summer Chum Salmon were counted (Table 1). Daily counts of summer Chum Salmon were very low for the first two weeks of weir operation, and the first Chinook Salmon was not seen until July 4 (Figure 2). The peak for both species occurred on July 19, which was relatively late. The highest 3-day counts of Chinook and summer Chum salmon occurred July 17–19, with 785 Chinook (55.1% of total run) and 823 summer Chum salmon (32.5% of total run) passing through the weir.

Only 90 Chinook Salmon were sampled for age, sex, and length in 2021, falling short of the target of 220–240 fish. There were no issues with high-water temperatures exceeding thresholds for safe fish handling, however, sampling was suspended from July 9 at 2000 hours to July 12 at 1130 hours due to flooding. Sample sizes were not only below target ranges but were not distributed in proportion to the roughly equal passage numbers in each tertile. Of the 90 samples, 1 fish could not be aged at all and another 16 fish could only be aged in the marine portions of the scales due to scale regeneration. Because nearly all Alaska Chinook Salmon spend one year in fresh water, we assumed the 16 fish with only marine ages also had one freshwater year. Therefore, the effective sample size was 89 fish (Table 2). Escapement age-sex proportions could not be estimated prior to the mid-point of the run because of the small sample size during that period. Therefore, the overall sample age-sex proportions did not adequately represent East Fork Andreafsky River Chinook salmon escapement in 2021. The sample had 68.5% males and 31.5% females. The age composition of the sample was 41.6% age-1.2 (37 males), 1.1% age-2.1 (1 male), 38.2% age-1.3 (22 males, 12 females), and 19.1% age-1.4 fish (1 male, 16 females; Table 3). The average length of males was shorter than females in each age class. Age-1.3 males were 118 mm shorter than their female counterparts. The largest fish were age-1.4 females, averaging 724 mm. The average length of all 89 Chinook Salmon sampled was 563 mm (Table 4).

The summer Chum Salmon sample of 237 fish was within the target range of 220 to 240 fish. Of the fish sampled, 13 could not be aged due to various sampling errors, leaving an effective sample of 224 fish. Eight of 122 samples taken during the June 18 to July 15 sampling period were unusable, 3 of 75 samples taken July 16–19 were unusable, and 2 of 40 samples taken during the July 20–28 sampling dates were unusable. Only 17% of all usable samples were taken in the final tertile of the run, so samples were instead grouped into two strata, before and after the mid-point of the run (July 18). Nine of 153 samples were unusable during the June 18 to July 17 sampling dates. Four of 84 samples were unusable during the July 18 to July 28 sampling dates. Two-thirds of the samples were taken during the first half of the run, leaving 80 viable samples to cover the 1,312 fish that returned in the second half (Table 5). The estimated sex composition of the 2021 escapement was 44.6% male and 55.4% female. The estimated age composition was 0.9% age 0.2 (1 male, 1 female), 86.2% age 0.3 (85 males, 108 females), 10.3% age 0.4 (10 males, 13 females), and 2.7% age 0.5 (2 females; Table 6). Except for age-0.2 fish, male summer Chum Salmon were larger than females in each age class. Age-0.2 males averaged 480 mm and age-0.5 males averaged 564 mm; females ranged from 487 mm (age-0.3; the single age-0.2 female in the sample was larger) to 528 mm (age-0.4; the two age-0.5 females in the sample were smaller) on average (Table 7).

Other Species Counts and Environmental Data

Counts of other salmon passing the East Fork Andreafsky weir in 2021 included 7,030 Pink Salmon and 241 Sockeye Salmon. Neither of these should be considered complete escapement counts as Pink Salmon can pass freely through the weir pickets and the project timing does not allow for complete enumeration of the Sockeye Salmon run. Other incidental counts included 3,544 Whitefish (almost exclusively Humpback and Broad) and 115 Northern Pike (Table 1).

Water depth at the weir fluctuated between 0.94 and 1.55 m during 2021. Daily average water temperatures at the weir ranged from 9.6 to 17.4ºC (Table 8), which was significantly cooler than in 2019, when the temperature ranged from 14.0 to 21.8ºC. Seasonal maximum temperatures of 17°C occurred on 3 days during weir operation, July 18–20. These temperatures did not exceed guideline thresholds for safe fish handling in Alaska (Shink 2020). Air temperatures ranged from a low of 2ºC to a high of 28ºC. The precipitation gauge at the weir camp recorded 75.7 mm of rain during the season, with a single day high of 10.8 mm on July 5.

Discussion

Chinook and summer Chum salmon passed the East Fork Andreafsky weir in low numbers during the 2021 season, missing the lower bounds of the escapement goals by 675 and 37,466 fish, respectively (JTC 2021). The low Chinook Salmon numbers were not altogether unsurprising, with drainage-wide forecasts for the year predicting poor returns below the goal. The abysmal return of summer Chum Salmon, however, was unexpected since the pre-season forecast, while below-average, predicted that escapements in the drainage would still meet targets. Most summer Chum Salmon returned to the Andreafsky River in 2021 as age-0.3 (86.2%) and age-0.4 (10.3%) fish, matching historical trends (Conitz 2018, 2019; Larson 2019; Conitz and French 2020), and meaning they were from the 2016 and 2017 brood years. In 2020, when the summer Chum Salmon born in 2016 returned to the Yukon River as age-4 fish, they passed the Pilot Station sonar in the second smallest numbers since 1978 (JTC 2021). Although it is important to remember that the Andreafsky weir is downriver of Pilot Station and was not operated in 2020, the low number of age-5 summer Chum Salmon returning to the Andreafsky River in 2021 bolsters the assumption that the 2016 brood year suffered abnormally low survival. The 2017 brood year appears to have fared significantly better than 2016, but overall returns were still far below expected.

During the 2021 season, the Andreafsky weir appeared to be experiencing the widespread effects of a change in ocean conditions that heavily affected the 2016 and 2017 brood years (JTC 2021), and it is unclear when conditions will improve. In addition to changing ocean conditions, prolonged, high water temperatures in 2019 are thought to have directly affected spawning events and offspring survival (Conitz 2019; von Biela et al. 2020). Large numbers of unspawned summer Chum Salmon carcasses were observed floating down the lower reaches of the Andreafsky River in July 2019 (Conitz and French 2020). The full scope of the heat-related mortalities during 2019 remain unknown, but they were widespread throughout Western Alaska and other parts of the state (Conitz and French 2020; Westley 2020; von Biela et al. 2020). If warming water conditions are a primary driver of the low returns, it is reasonable to expect future years, at least in the near term, may have exceptionally poor returns as well.

In 2021, newly implemented water temperature monitoring equipment allowed for a closer examination of temperature trends throughout the season, and guidance from Shink (2020) was used to modify the operational plan for the weir to alleviate heat stress as much as possible. However, this guidance was not needed in 2021 due to an average water temperature of 12.6°C during July 1–28, 2021, which was markedly cooler than 2019, where water temperatures averaged 17.7°C over the same period. Similarly, the high in 2021 was 17.3°C on July 19 compared to the 2019 high of 23.5°C on July 7. At no point were weir operations altered due to high water temperatures in 2021, which would have happened if the average water temperature exceeded 17°C for 3 consecutive days (Shink, 2020).

Collecting biological samples according to a stratified sampling design, and roughly in proportion to run timing, was accomplished for summer Chum Salmon but not Chinook Salmon in 2021. The primary reason for distributing biological samples through the run is to ensure they are not biased by variation in age and sex composition as the run progresses (Conitz and French 2020). For example, males often lead the migration in Chum Salmon (Salo 1991), so disproportionate sampling early in the season could result in overestimating the proportion of males in the run. In 2021, the total sample size goal was met for summer Chum Salmon, and the samples were distributed evenly across early and late portions of the run. However, the samples of Chinook Salmon were too heavily weighted towards the latter part of the run and were too small to provide valid inferences about the population. These issues mirrored the 2019 season, when the sampling goal for Chinook Salmon was also missed, with the samples that were collected being too heavily weighted towards the early part of the run.

The sampling goal was missed primarily due to insufficient time spent trapping salmon during their peak hours of return. In part, this was due to high water at the site, which necessitated the temporary removal of a section of the weir and the cessation of trapping for 63.5 hours from July 9 to 13. Ultimately, the trap was utilized for 350.6 hours during the season, equating to 37.2% of the time the weir was fully operational. In most years, high daily fish passage typically requires the sampling trap to be operated for only a small portion of each day, allowing for fish to pass through the weir nearly continuously, maintaining a more natural migration and reducing exposure to heat stress. Hourly passage is roughly even on an interannual basis. However, within a given year, Chinook Salmon often pass the weir during "peak" hours, which remain consistent throughout the season. In 2021, 71% of Chinook Salmon passed the weir during a 6-hour block from 0100 to 0700 hours, and the mid-season trapping schedule failed to cover that period adequately. Trapping hours varied throughout the season, allowing the trap to cover an even spread of times throughout the day, but they were not adjusted quickly enough to account for peak passage (Figure 3). In 2021, 74% of the Chinook Salmon run passed the weir in just five days, highlighting the need for rapid changes to the trapping schedule or the conversion to a manual observation weir during the suspected peak of the run (Figure 4). After a slow start to Chinook sampling, changes in crew shift scheduling allowed for more targeted capture of Chinook Salmon, leading to the collection of 57 scales (63% of the total) over the last 7 days of weir operation. In comparison, only 17 Chinook Salmon were sampled prior to the midpoint of the run, on July 18 (Table 2).

Interestingly, a relatively large number of age-1.2 jacks were sampled despite collecting most of the Chinook Salmon samples in the latter half of the run. The presence of higher numbers of young males suggests that juveniles of the upcoming cohort survived the transition from freshwater to marine conditions, so the returns next year may be better than in 2021. At the same time, the large presence of young males may indicate that the age distribution on the East Fork of the Andreafsky River is continuing to become gradually younger. The 2021 season was the 7th season in a row where no age-7 Chinook Salmon were sampled, and the last 10-year period that can be assessed with complete annual samples, 2009–2018, saw an average of only 0.2% age-7 fish. Meanwhile, the proportion of age-4 fish has steadily expanded from a 10-year average of 15.8% (1985–1994) to 33.7% (2009–2018).

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Table 1. — Daily and cumulative escapement estimates of Chinook, summer Chum, and Pink salmon, and daily estimates of other species, at the East Fork Andreafsky River weir, Alaska, 2021.

Date[f]	Chinook Daily	Chinook Cum	Chum Daily	Chum Cum	Pink Daily	Pink Cum	Sockeye Daily	Whitefish Daily	Pike Daily
Jun 18	0	0	0	0	0	0	0	19	9
Jun 19	0	0	8	8	0	0	0	10	15
Jun 20	0	0	2	10	0	0	0	3	7
Jun 21	0	0	3	13	0	0	0	13	3
Jun 22	0	0	1	14	0	0	0	31	6
Jun 23	0	0	1	15	0	0	0	12	2
Jun 24	0	0	3	18	0	0	0	3	2
Jun 25	0	0	3	21	0	0	0	11	2
Jun 26	0	0	2	23	0	0	0	7	1
Jun 27	0	0	1	24	0	0	0	5	1
Jun 28	0	0	0	24	0	0	0	1	1
Jun 29	0	0	2	26	1	1	0	4	0
Jun 30	0	0	8	34	0	1	0	13	3
Jul 1	0	0	3	37	0	1	1	31	3
Jul 2	0	0	57	94	0	1	0	50	1
Jul 3	0	0	38	132	4	5	0	62	1
Jul 4	6	6	11	143	4	9	0	33	1
Jul 5	0	6	6	149	1	10	0	33	1
Jul 6	7	13	60	209	3	13	9	41	4
Jul 7	2	15	36	245	3	16	3	16	4
Jul 8	5	20	18	263	0	16	3	13	1
Jul 9	15	35	93	356	5	21	3	28	0
Jul 10	31	66	99	433	5	26	2	32	0
Jul 11	1	67	108	463	2	28	0	16	1
Jul 12	4	71	12	475	5	33	5	36	10
Jul 13	42	113	96	571	45	78	18	163	7
Jul 14	95	208	127	698	71	149	29	307	6
Jul 15	52	260	50	748	113	262	16	157	1
Jul 16	87	347	178	926	279	541	24	261	4
Jul 17	214	561	296	1,222	529	1,070	26	260	1
Jul 18	179	740	173	1,395	384	1,453	17	326	2
Jul 19	392	1,132	354	1,749	1,979	3,432	26	544	0
Jul 20	187	1,319	71	1,820	929	4,360	12	367	4
Jul 21	36	1,355	54	1,874	211	4,571	8	142	3
Jul 22	9	1,364	53	1,927	116	4,687	1	118	3
Jul 23	4	1,368	62	1,989	142	4,829	6	136	3
Jul 24	3	1,371	32	2,021	94	4,923	7	53	1
Jul 25	4	1,375	85	2,106	823	5,746	10	44	0
Jul 26	14	1,389	134	2,240	436	6,182	1	64	1
Jul 27	6	1,395	92	2,332	297	6,479	5	31	0
Jul 28	30	1,425	202	2,534	551	7,030	11	49	0
Total count	1,425		2,534		7,030		241	3,544	115
Est. 6/15-6/17[g]	0		0						
Est. 7/29-8/10[h]	29		150						
Total estimate	1,454		2,684						
95% HPDI[i]	1,435-1,478		,596-2,772						

[f] Weir counts for July 10–11 were estimated as less than 6 hours of counts were available for each day. If the estimate was less than the partially observed count, the observed count was used.
[g] Estimated from June 15 (based on historical run timing) until first day of weir operation, June 18.
[h] Estimated from last day of weir operation until August 10 (based on historical run timing).
[i] 95% Highest posterior density estimate (Bayesian equivalent to confidence interval)

Table 2. — Chinook Salmon age-sex-length sample sizes and proportions at the East Fork Andreafsky River weir, 2021. With the run divided into tertiles and into two periods of approximately equal fish passage proportions.

Period dates	Passage counts	Proportion of passage	Number of samples	Proportion of sample
Tertials				
Jun 28 – Jul 16[a]	347	0.25	10 (5)	0.11
Jul 17 – Jul 19	775	0.55	18 (3)	0.20
Jul 20 – Jul 28	296	0.21	61 (8)	0.69
Totals	1,425		89 (16)	
Two periods				
Jun 28 – Jul 18[a]	740	0.52	22 (8)	0.25
Jul 19 – Jul 28	685	0.48	67 (8)	0.75
Totals	1,425		89 (16)	

[a] No sampling occurred July 10–11 due to flooding.

Table 3. — Chinook Salmon sample age and sex composition and 95% confidence intervals (CI) at the East Fork Andreafsky River weir, 2021.

Grouped by	Sex	Age	Number of fish	Proportion	95% CI Lower bound	95% CI Upper bound	95% C I Width
Grouped by sex & age	M	1.2	37	41.6%	31.3%	51.8%	20.5%
	M	2.1	1	1.1%	<0.1%	3.3%	3.3%
	M	1.3	22	24.7%	15.8%	33.7%	17.9%
	M	1.4	1	1.1%	<0.1%	3.3%	3.3%
	F	1.3	12	13.5%	6.4%	20.6%	14.2%
	F	1.4	16	18.0%	10.0%	26.0%	16.0%
Grouped by age		1.2	37	41.6%	31.4%	51.8%	20.5%
		2.1	1	1.1%	<0.1%	3.3%	3.3%
		1.3	34	38.2%	28.1%	48.3%	20.2%
		1.4	17	19.1%	10.9%	27.3%	16.3%
Grouped by sex	M		61	68.5%	58.9%	78.2%	19.3%
	F		28	31.5%	21.8%	41.1%	19.3%

Table 4. — Chinook Salmon average length -at- age by sex at the East Fork Andreafsky River weir, 2021. The 89 fish that were aged had an average length of 563 mm with a standard error of 12 mm.

Age	Sex	Number of fish	Avg length (mm)	SE
1.2	M	37	466	7
2.1	M	1	390	–
1.3	M	22	554	9
1.4	M	1	610	–
1.3	F	12	672	7
1.4	F	16	724	13

Table 5. — Summer Chum Salmon age-sex-length sample sizes and proportions at the East Fork Andreafsky River weir, 2021. With the run divided into tertiles and into two periods.

Period dates	Passage counts	Proportion of passage	Number of samples	Proportion of sample
Tertials				
Jun 18 – Jul 15[a]	748	0.30	114	0.51
Jul 16–19	1,001	0.40	72	0.32
Jul 20–28	785	0.31	38	0.17
Totals	2,534		224	
Two periods				
Jun 28 – Jul 17[a]	1,222	0.48	144	0.64
Jul 18-28	1,312	0.52	80	0.36
Totals	2,534		224	

[a] No sampling occurred July 10 through July 11 due to flooding.

Table 6. — Summer Chum Salmon sample age and sex composition and 95% confidence intervals (CI) at the East Fork Andreafsky River weir, 2021.

Grouped by	Sex	Age	Number of fish	Proportion	95% C I Lower bound	95% C I Upper bound	95% C I Width
Grouped by sex & age	M	0.2	1	0.4%	<0.1%	1.2%	1.2%
	M	0.3	85	37.9%	31.5%	44.3%	12.7%
	M	0.4	10	4.5%	1.8%	7.2%	5.4%
	M	0.5	4	1.8%	0.1%	3.5%	3.5%
	F	0.2	1	0.4%	<0.1%	1.2%	1.2%
	F	0.3	108	48.2%	41.7%	54.7%	13.1%
	F	0.4	13	5.8%	2.7%	8.9%	6.1%
	F	0.5	2	0.9%	<0.1%	2.1%	2.1%
Grouped by age		0.2	2	0.9%	<0.1%	2.1%	2.1%
		0.3	193	86.2%	81.7%	90.7%	9.0%
		0.4	23	10.3%	6.3%	14.3%	8.0%
		0.5	6	2.7%	0.6%	4.8%	4.2%
Grouped by sex	M		100	44.6%	38.1%	51.1%	13.0%
	F		124	55.4%	48.9%	61.9%	13.0%

Table 7. — Summer Chum Salmon average length-at-age by sex at the East Fork Andreafsky River weir, 2021. The 224 fish that were aged had an average length of 503 mm with a standard error of 2 mm.

Age	Sex	Number of fish	Avg length (mm)	SE
0.2	M	1	480	–
0.2	F	1	494	–
0.3	M	85	514	4
0.3	F	108	487	3
0.4	M	10	552	11
0.4	F	13	528	10
0.5	M	4	564	25
0.5	F	2	503	8

Table 8. — Daily water level (in meters, as measured from streambed to water surface at fish passage chute; not calibrated to shoreline elevation), turbidity, and temperature (C), and daily air temperature (C) at the East Fork Andreafsky River weir, Alaska, 2021.

Date	Water Level AM	Water Level PM	Turbidity (visual)	Water Temp Average	Water Temp Minimum	Water Temp Maximum	Air Temp Minimum	Air Temp Maximum
Jun 18	1.00	1.10	Clear	14.5	12.7	15.8	12	20
Jun 19	1.22	1.19	Clear	14.4	13.0	15.6	7	17
Jun 20	1.16	1.14	Moderate	14.1	13.3	14.9	8	18
Jun 21	1.13	1.11	Clear	14.1	12.8	15.6	11	20
Jun 22	1.10	1.11	Clear	14.8	14.1	15.4	10	22
Jun 23	1.11	1.08	Clear	14.4	13.7	15.3	9	19
Jun 24	1.07	1.04	Clear	13.8	13.1	14.6	10	20
Jun 25	1.04	1.04	Clear	13.6	12.6	14.5	10	18
Jun 26	1.01	1.03	Clear	13.5	12.9	14.1	8	16
Jun 27	1.01	1.02	Clear	12.6	12.3	13.4	9	14
Jun 28	1.05	1.05	Clear	12.2	11.7	12.8	9	17
Jun 29	1.02	1.02	Clear	12.2	11.8	12.5	9	15
Jun 30	0.99	0.97	Clear	12.2	11.5	13.3	10	21
Jul 1	0.97	0.97	Clear	13.4	12.6	14.3	12	18
Jul 2	0.94	0.94	Clear	13.5	12.9	14.2	10	16
Jul 3	0.95	0.97	Clear	12.7	12.0	13.7	8	14
Jul 4	0.98	0.99	Clear	11.7	11.4	12.4	9	14
Jul 5	1.02	1.08	Moderate	10.9	10.8	11.3	9	13
Jul 6	1.15	1.18	Moderate	10.5	10.1	11.3	9	13
Jul 7	1.22	1.22	Moderate	10.7	10.3	11.2	6	12
Jul 8	1.19	1.18	Moderate	10.4	9.9	10.9	8	12
Jul 9	1.24	1.37	High	10.2	9.6	11.0	8	13
Jul 10	1.55	1.50	High	ND	9.9	ND	6	19
Jul 11	1.40	1.30	High	ND	ND	ND	5	18
Jul 12	1.27	1.23	Moderate	ND	ND	14.1	2	22
Jul 13	1.19	1.19	Clear	13.3	12.3	14.8	5	17
Jul 14	1.14	1.12	Clear	13.9	13.4	14.8	8	17
Jul 15	1.09	1.09	Clear	13.6	12.6	15.2	9	17
Jul 16	1.07	1.06	Clear	13.5	12.8	14.5	6	16
Jul 17	1.04	1.02	Clear	13.9	13.3	14.6	11	20
Jul 18	1.01	1.00	Clear	14.9	13.3	17.3	11	28
Jul 19	0.99	0.98	Clear	16.5	15.6	17.4	11	23
Jul 20	0.99	1.00	Clear	15.7	14.6	17.2	11	16
Jul 21	1.12	1.13	Moderate	14.3	13.7	15.0	11	16
Jul 22	1.10	1.06	Clear	13.9	13.0	14.9	8	14
Jul 23	1.04	1.03	Clear	13.2	12.4	14.5	4	11
Jul 24	1.01	0.99	Clear	11.9	11.3	12.3	8	16
Jul 25	0.98	0.98	Clear	11.9	11.4	12.4	10	14
Jul 26	0.97	0.96	Clear	11.8	11.6	12.3	10	12
Jul 27	0.98	1.01	Clear	11.3	11.1	11.7	10	13
Jul 28	1.09	1.17	Moderate	11.0	10.8	11.1	11	13
Minimum	1.55	1.50		9.9	1.6	10.9	12	28
Maximum	0.94	0.94		16.5	15.6	19.8	2	11
Average	1.09	1.09		12.9	11.8	14.2	9	17

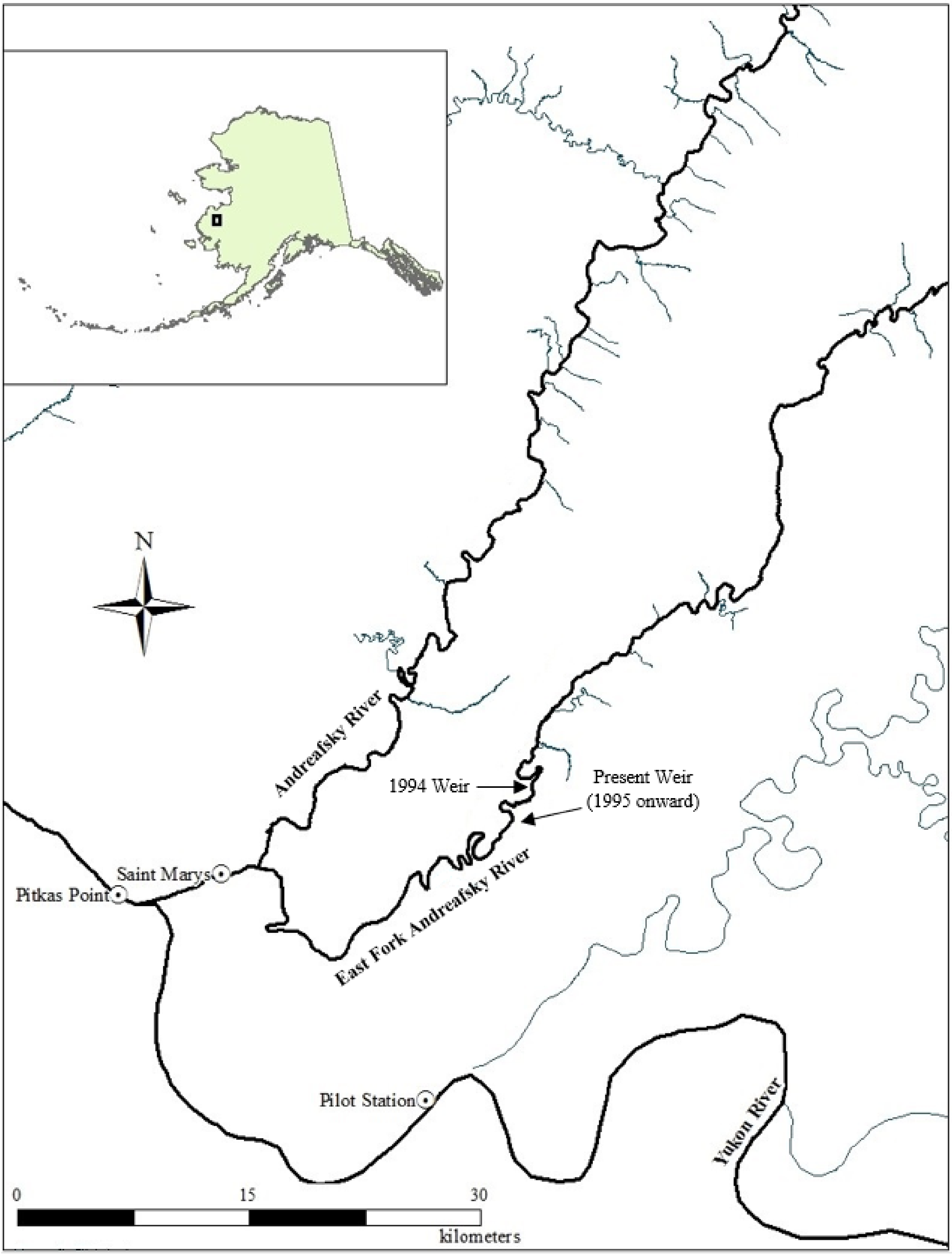

Figure 1. — Present weir location on the East Fork Andreafsky River, Alaska, 1995–2021. Note: the 1994 site was 2.4 rkm upstream of the current weir location (Tobin and Harper 1996).

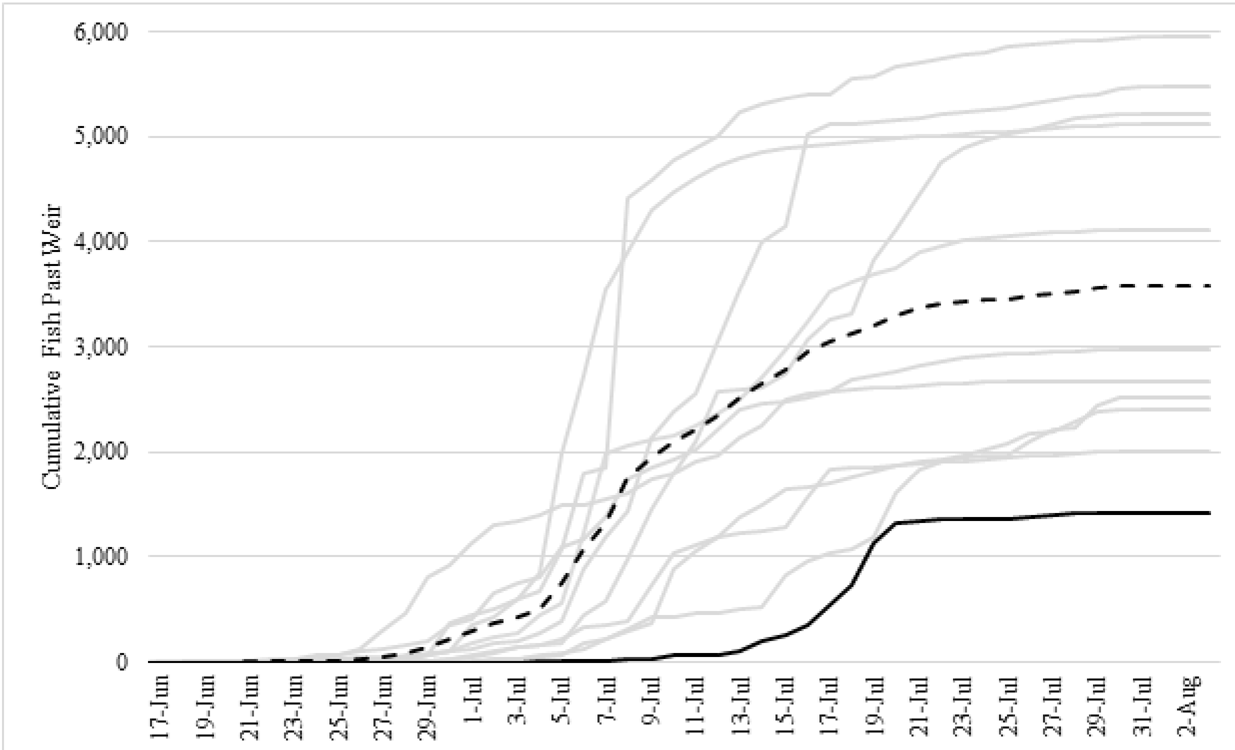

Figure 2. — Cumulative Chinook Salmon passage 2010-2021. The 2021 season is shown in black. All other years are shown in gray, with the 10-year average shown as a dashed line.

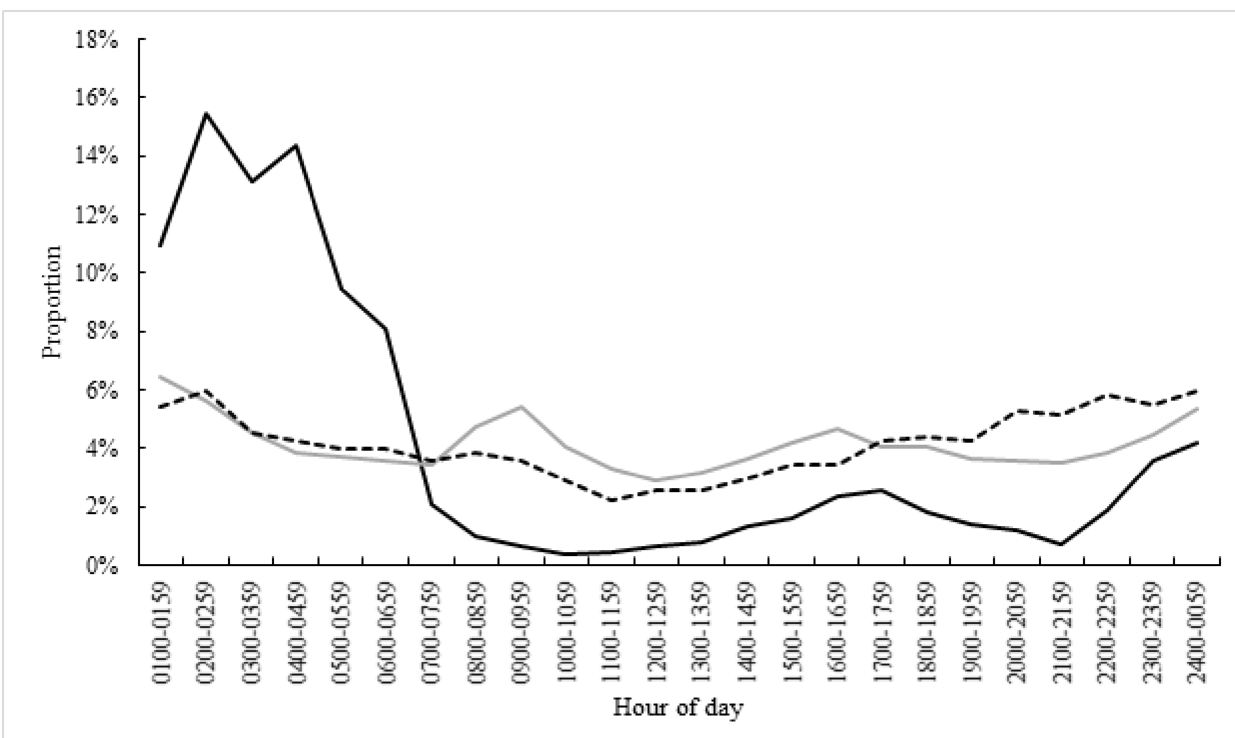

Figure 3. — Proportional Chinook Salmon passage by hour during the 2021 season (solid black line) and the average of the past 6 years of data (solid grey line). The distribution of hours spent trapping during 2021 (dashed black line) is also shown.

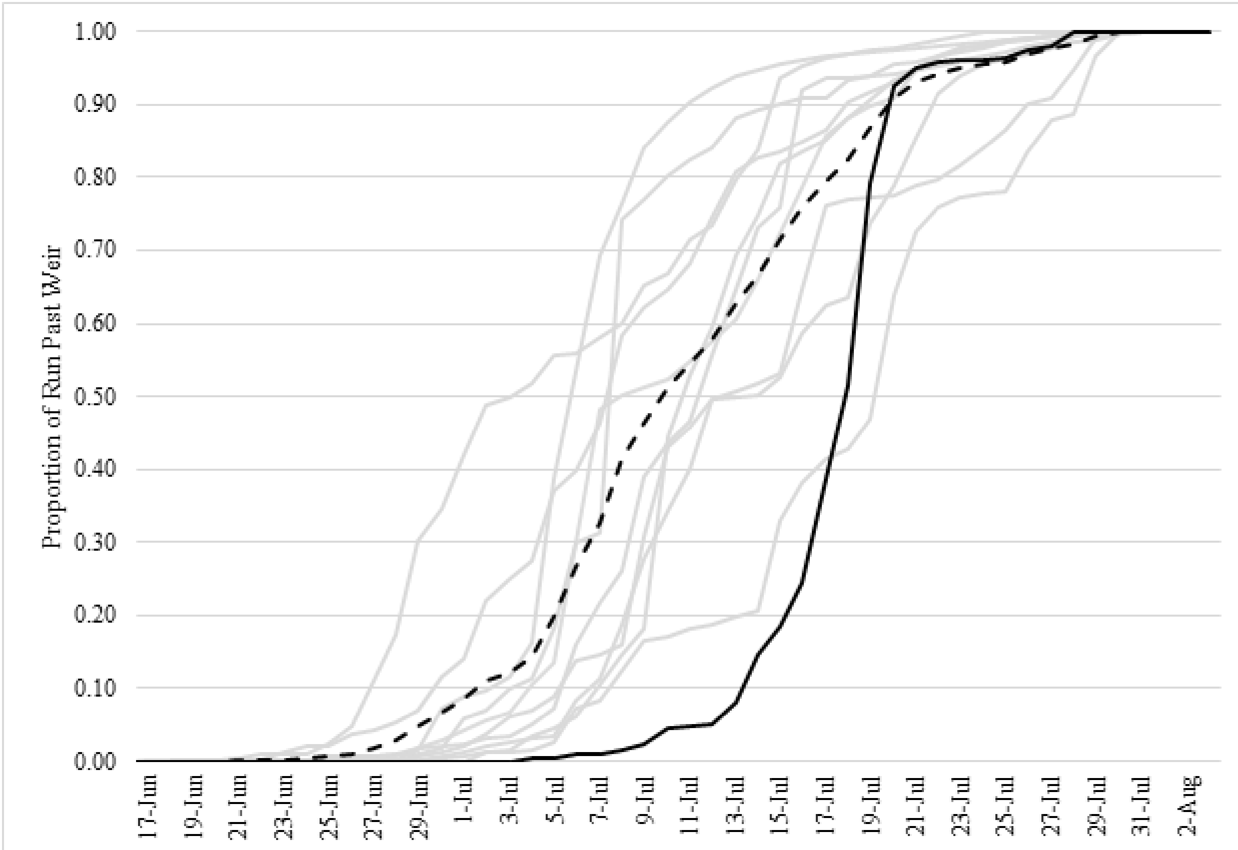

Figure 4. — Proportional Chinook Salmon passage 2010-2021. The 2021 season is shown in black. All other years are shown in grey, with the 10-year average shown as a dashed line.